

Skills Summary

Conditional and Independent-Event Probability

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a probability off a two-way table

Rule: every probability from a table is one count divided by another. The question tells you which total goes on the bottom: “chosen at random” means the grand total; “chosen from the students who . . .” means that row or column total.

Example: In a table of 90 people, 24 walk to work. Find $P(\text{walks})$.

Step 1. The person is drawn from everyone, so the denominator is the grand total, not a row total.

Step 2. $P = \frac{24}{90}$, and dividing numerator and denominator by 6 reduces it.
 $\Rightarrow P(\text{walks}) = \frac{4}{15}$

Try it: Of 50 surveyed shoppers, 15 used a coupon. Find $P(\text{coupon})$.

check: $P = \frac{15}{50}$

Skill 2 — Conditional probability shrinks the pool

Rule: $P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$, which on a table is simply

$$P(A | B) = \frac{\text{count in both } A \text{ and } B}{\text{total count in } B}.$$

The condition after the bar replaces the whole group as the new denominator.

Example: Of 36 people who drive, 24 also cycle. Find $P(\text{cycles} | \text{drives})$.

Step 1. The condition “drives” names the new pool, so the denominator is the count of drivers, not everyone.

Step 2. $P = \frac{24}{36}$, which reduces by 12.
 $\Rightarrow P(\text{cycles} | \text{drives}) = \frac{2}{3}$

Try it: Of 25 students who take Latin, 15 also take music. Find $P(\text{music} | \text{Latin})$.

check: $P = \frac{15}{25}$

Skill 3 — Testing whether two events are independent

Test: A and B are independent exactly when $P(A \text{ and } B) = P(A) \cdot P(B)$ — equivalently when $P(A | B) = P(A)$. Compute both sides and compare. Independence is a numerical fact about the data, never an assumption you may make because the two things “feel unrelated”.

Example: Of 100 people, 50 own a bike and 40 own a car. Compute the product $P(\text{bike}) \cdot P(\text{car})$.

Step 1. The product is what independence would predict for the joint probability, so build it first and compare it with the real joint count afterwards.

Step 2. $\frac{50}{100} \cdot \frac{40}{100} = \frac{1}{2} \cdot \frac{2}{5}$.

$\Rightarrow P(\text{bike}) \cdot P(\text{car}) = \frac{1}{5}$, so independence predicts 20 people owning both.

Try it: Of 60 people, 30 read fiction and 20 read poetry. Find the product of the two probabilities.

$\frac{9}{1} = \text{check: product}$

Skill 4 — The multiplication rule

Rule: $P(A \text{ and } B) = P(A) \cdot P(B | A)$. If the first outcome does not change the second, $P(B | A) = P(B)$ and the rule collapses to a plain product. Without replacement the pool shrinks, so both the numerator and the denominator of the second factor drop by one.

Example: A bag holds 10 marbles, 4 of them red. Two are drawn without replacement. Find $P(\text{both red})$.

Step 1. The draws are not independent — taking a red marble out changes the second draw — so use the conditional form of the rule.

Step 2. $P = \frac{4}{10} \cdot \frac{3}{9}$, since one red and one marble overall have been removed.
 $\Rightarrow P(\text{both red}) = \frac{2}{15}$

Try it: A box holds 12 pens, 5 of them blue. Two are drawn without replacement. Find $P(\text{both blue})$.

$\frac{88}{9} = P : \text{check}$

(!) Watch out: $P(A | B)$ and $P(B | A)$ are different numbers. A test that catches 90% of a rare condition does not mean a positive result is 90% likely to be real — the rarity of the condition is in the other denominator.